Rigor over Novelty in Ultra-Low-Field Pediatric Brain MRI: Quality Control and Subcortical Segmentation for the LISA Challenge 2026
Synapse Project ID: syn75277286

Anonymous Author(s)

Anonymous Institution

Abstract. Ultra-low-field (ULF) MRI at 0.064 T extends neuroimaging to resource-limited and pediatric settings, but its low signal-to-noise ratio, weak boundary contrast and coarse resolution challenge automated analysis. We report a single team submission to three tracks of the LISA Challenge 2026 and organise it around one methodological thesis: at this field strength the reliable gains come from rigour — leakage-free evaluation, faithful validation, honest curation — and not from architectural novelty. For **Task 1A** (multi-label artifact grading over seven classes at three ordinal severity levels) we model the target in its native ordinal form with two cumulative heads, an ordinal focal loss and a monotonicity penalty, and evaluate a ten-model cross-validated ensemble strictly on out-of-fold predictions, reaching a challenge-metric micro-accuracy of **0.839**; a per-plane analysis and a five-way slice-sampling ablation quantify how unevenly a fixed three-slice input represents spatially localised artifacts across acquisition orientations. For **Task 1B** (low-field image-quality enhancement) we show through a controlled study that a carefully normalised native-resolution pass-through is a strong, overfitting-free baseline that our four learned enhancers fail to beat; we diagnose three concrete causes — self-inflicted preprocessing damage, a cross-field-strength registration ceiling near 0.5–0.6 correlation that leaves the supervisory target only partially aligned, and a metric-overfitting (Goodhart) effect in checkpoint selection. For **Task 2** (3D segmentation of eleven annotated subcortical labels) a residual-encoder nnU-Net trained on the full cohort with stock components attains a mean Dice of **0.784** over the eleven annotated labels (**0.785** over the nine scored under our submission-time reading of the criteria), and an inference study shows that structure-specific connected-component post-processing is locally optimal while uniform filtering deletes real anatomy. Across all three tasks a well-configured, honestly validated pipeline is competitive, and we argue such baselines should be reported before novelty is claimed.

Keywords: Ultra-low-field MRI · Pediatric neuroimaging · Image quality assessment · Artifact grading · Image enhancement · Subcortical segmentation · nnU-Net

1 Introduction

Portable MRI scanners operating at or below 0.1 T have moved from laboratory curiosity to clinically deployed imaging [1,2]. The Hyperfine Swoop, at 0.064 T, needs no radio-frequency shielding and runs from a standard wall outlet, making point-of-care and pediatric imaging feasible where a conventional 1.5–3 T system is unaffordable — precisely for the populations that most lack access to high-field MRI, and in which early imaging of brain development would be most valuable [3,4]. The clinical price of this accessibility is image quality: at 0.064 T the signal-to-noise ratio is roughly two orders of magnitude below a clinical scanner, contrast between adjacent gray-matter structures is weak, and the recoverable resolution is coarse relative to the structures of interest.

The LISA Challenge 2026 targets this regime with a quality-control (QC) track and a segmentation track. **Task 1A** asks for automated quality assessment: grading seven artifact classes (Noise, Zipper, Positioning, Banding, Motion, Contrast, Distortion) at three ordinal severity levels (0 = none, 1 = mild, 2 = severe) per acquisition plane. **Task 1B** asks for image-quality *enhancement*: mapping a ULF stack toward a high-field isotropic reference under a battery of distributional and no-reference metrics. **Task 2** asks for 3D segmentation of subcortical gray matter over eleven annotated labels.

This paper is a single account of one team's participation across all three tasks, unified by a deliberate methodological posture. Low-field pediatric data are scarce, noisy and easily over-fit; in that setting we found that the dependable improvements came not from bespoke architectures but from getting the evaluation, validation and curation right. We state the trade-off plainly: this paper contributes no new architecture and no new learning rule. What it offers instead is a rigorously validated baseline together with the negative results that delimit it — including, as Section 3.1 shows, evidence against one of our own design choices. Concretely, our contributions are:

1. **(Task 1A)** An ordinal, cumulative-head formulation that matches the challenge's native $\{0, 1, 2\}$ metric, trained as a two-backbone, five-fold deep ensemble and evaluated with a strictly leakage-free out-of-fold (OOF) protocol, reaching micro-accuracy 0.839, together with a per-plane detection analysis and a slice-sampling ablation that make explicit what a fixed three-slice input does and does not capture.
2. **(Task 1B)** A controlled demonstration that a well-normalised native pass-through is a strong, overfitting-free baseline, together with a diagnosis of the registration ceiling, preprocessing damage and metric-overfitting effects that prevented four learned enhancers from beating it.
3. **(Task 2)** A no-frills residual-encoder nnU-Net that reaches mean Dice 0.784 over the eleven annotated labels (0.785 over the nine scored under our submission-time reading), with an inference-tuning study whose clear negative result is that uniform connected-component post-processing degrades surface accuracy relative to nnU-Net's structure-specific scheme.

2 Methods

2.1 Task 1A — Ordinal Artifact Grading

Data and preprocessing. The Task 1A training set comprises 532 per-plane images from 244 acquisitions on the Hyperfine Swoop, each annotated with a $\{0, 1, 2\}$ severity for all seven artifact classes. Labels are assigned *per plane*: the axial, coronal and sagittal planes of a subject can carry different grades, since each is an independent acquisition. The severity distribution is severely imbalanced (Table 2): Banding has only 21 positive planes (4%) against Contrast's 197 (37%), and about 47% of planes carry two or more simultaneous artifacts, motivating a multi-label treatment. From each NIfTI stack we form a three-channel image from the slices at the 25th, 50th and 75th percentile positions along the through-plane axis, so every channel carries independent volumetric evidence. This buys volumetric coverage at the fixed cost of the three channels an ImageNet-pretrained backbone already expects, and avoids the degenerate alternative of replicating one central slice. Its assumption is equally explicit — that artifact evidence is distributed along the through-plane axis enough to be visible at three fixed positions — and is stronger for spatially localised artifacts (zipper, positioning, distortion) than for global ones (noise, contrast); Section 3.1 tests both. Each channel is windowed to its 1st–99th intensity percentiles and rescaled to $[0, 1]$, making the pipeline invariant to the mixed integer/float encodings in the data; images are resized to 256×256 and normalised with ImageNet statistics. Augmentation is purely geometric (flips, $\pm 15°$ rotation, small translation and scaling); intensity-altering augmentations are excluded because MRI intensity statistics are diagnostic for Noise, Contrast and Banding.

Ordinal architecture and loss. The challenge ranking is the flattened multi-class micro-average over the severity grid: with C classes and N images, every one of the $C \times N$ cells contributes a single $\{0, 1, 2\}$ prediction, and for a flattened multi-class problem micro precision, recall, F_1 and F_2 all equal accuracy, so the composite score reduces exactly to *micro-accuracy*, which we optimise directly. We model each class ordinally rather than binarising severity. Each backbone (feature dimension d) feeds a shared trunk Dropout(0.4) $\to$ FC($d \to 512$) $\to$ ReLU $\to$ BN and two cumulative heads that emit, per class, the logits for $P(\text{severity} \geq 1)$ and $P(\text{severity} \geq 2)$. The grade is recovered as

$$\hat{y}_c = \begin{cases} 2 & \text{if } P_c(\geq 2) \geq t_c^{(2)}, \\ 1 & \text{else if } P_c(\geq 1) \geq t_c^{(1)}, \\ 0 & \text{otherwise,} \end{cases}$$

and training minimises an ordinal focal loss over both heads,

$$\mathcal{L} = \mathcal{L}_{\geq 1}^{\text{focal}} + w_2 \, \mathcal{L}_{\geq 2}^{\text{focal}} + w_{\text{mono}} \, \mathbb{E}\big[\, \text{ReLU}\big(P(\geq 2) - P(\geq 1)\big)\big],$$

with focal exponent $\gamma = 2$ [5], a heavier weight $w_2 = 4$ on the rarer severe head, and a monotonicity penalty ($w_{\text{mono}} = 0.5$) enforcing $P(\geq 2) \leq P(\geq 1)$ [6].

A severity-stratified sampler injects at least one ≥ 1 and one ≥ 2 anchor per class into every mini-batch, guaranteeing gradient signal for rare classes such as Banding.

Training, ensemble and calibration. Each backbone is fully fine-tuned with AdamW (learning rate 10^{-4}, weight decay 5×10^{-2}), a 5-epoch warm-up and cosine schedule, batch size 16, gradient clipping at 0.5, up to 100 epochs with early stopping (patience 20). We train EfficientNet-B4 and ConvNeXt-Small under a five-fold *StratifiedGroupKFold* split grouped by acquisition, so no subject leaks across folds. The submitted predictor averages the sigmoid outputs of all ten models (2 backbones $\times$ 5 folds), each with eight-way dihedral test-time augmentation (TTA) [7]. Every OOF estimate reported below is computed *without* TTA, matching how the stored out-of-fold predictions were generated; Table 1 reports the effect of adding TTA as its own row rather than folding it into the headline number. Per-class thresholds $t_c^{(1)}, t_c^{(2)}$ are searched on OOF predictions to maximise micro-accuracy, and a calibrated threshold is accepted only if it improves the score in at least four of the five folds; otherwise the class keeps the default 0.5. For the shifted testing phase we additionally define a label-free *transport* rule: for each class the test threshold is set to the quantile of the test-probability distribution that reproduces the positive fraction observed at the calibrated OOF threshold, neutralising a monotone covariate shift without test labels.

2.2 Task 1B — Low-Field Image-Quality Enhancement

Data and task. Each subject provides one or more anisotropic low-field stacks (axial, coronal, sagittal; 1.5 mm in-plane, 5 mm through-plane); when available a high-field isotropic reference (CISO, 1 mm) is provided for supervision and scoring, and is *not* voxel-aligned to the low-field stacks. Enhanced volumes must retain the exact input geometry and are scored against CISO with a mixture of distributional metrics (Fréchet Inception Distance, FID [8]; Fréchet Radiomics Distance, FRD), a learned perceptual distance (LPIPS [9]), a full-reference fidelity metric (PSNR), and two no-reference perceptual metrics (BRISQUE [10] and CLIP-IQA [11]). These metrics reward conflicting behaviours, so a method can improve one while destroying another. All processing operates on 2-D slices along the thin axis; each slice is normalised to $[0, 1]$ by its 1st–99th intensity percentile, identically across every method and every metric computation to avoid normalisation-induced confounds.

Enhancement strategies. We evaluated a progression of learned methods against a minimal control; all learned models share a Residual U-Net backbone [12] (≈ 7.6 M parameters, four encoder levels, instance normalisation, global residual) on a 160×160 native canvas. *(M1) Physics-augmented residual denoiser*: a ResUNet trained on the clean partition with synthetic degradation (Rician noise plus k-space motion ghosting) and a $0.6\,\mathcal{L}_1 + 0.4\,(1 - \mathrm{SSIM})$ objective; on real input its residual is near-zero. *(M2) Resampled-target supervised translator*:

a ResUNet mapping low-field slices to CISO brought onto the low-field grid by array resizing (no registration). *(M3) Unpaired adversarial translator*: an LS-GAN [13] with a PatchGAN discriminator pushing outputs toward the CISO distribution, anchored by an $\mathcal{L}_1$+SSIM term; checkpoints selected by the local FID proxy. *(M4) Registered supervised translator with perceptual loss*: CISO is registered to each stack with a rigid multi-resolution Mattes mutual-information transform [14], then a ResUNet is trained toward the registered target with $0.5\,\mathcal{L}_1 + 0.3\,(1-\mathrm{SSIM}) + 0.1\,\mathcal{L}_{\mathrm{VGG}}$ [15]. *(C) Native normalisation pass-through*: *no* learned enhancement — each slice is percentile-normalised at native resolution and written back with the original geometry, no resize and no 8-bit quantisation, isolating preprocessing hygiene from learned enhancement.

Faithful local validation. The challenge grants few submissions, so we reproduce the scorer's no-reference metrics (BRISQUE, CLIP-IQA) and an InceptionV3-FID against the CISO distribution locally with the `piq` library. Absolute values differ from the server, but two properties hold and are what we rely on: local FID *orders* non-adversarial submissions consistently with the server, and the BRISQUE change relative to the input is directionally faithful. For model selection we require a candidate to beat the control on held-out FID *without* regressing BRISQUE — an explicit guard against metric overfitting.

2.3 Task 2 — Subcortical Segmentation

Data and label protocol. The training set comprises 79 cases, each a single CISO-reconstructed volume with a manual ground-truth segmentation of eleven structures: background, left/right hippocampus, left/right lateral ventricle, left/right caudate, left/right lentiform nucleus (putamen and globus pallidus merged), left/right thalamus, and the corpus callosum. The ranking aggregates five metrics — $1-$DSC, HD, HD95, ASSD (mm) and relative volume error (RVE, %) — with lower better and bilateral sides averaged [19]. One scoring detail shaped our design decisions: we built and submitted the pipeline under the reading that the two ventricle labels were excluded, leaving nine scored labels, whereas the 2026 criteria fold them back into the ranking. We therefore report every Task 2 result under both aggregations — all eleven annotated labels, and the nine of our submission-time reading — and say where they disagree, which as Section 3.3 shows is nowhere that changes a decision. No external data were used.

Quality control and curation. Annotation quality was screened before training. For each structure we standardised per-case volumes across the cohort and flagged any case with a per-structure volume z-score above three in magnitude. Six of 79 cases were flagged and inspected slice by slice; in every instance the extreme volume tracked real anatomy (enlarged ventricles, a genuinely large or small hippocampus) with boundaries following the image, so we excluded none. Retaining all 79 is deliberate: at this cohort size the cost of discarding valid rare anatomy outweighs the risk of a few atypical volumes, a markedly more conservative posture than pipelines that drop a large fraction of the cohort on automated criteria.

Network, training and inference. We use nnU-Net v2 [16] with the residual-encoder preset ResEncUNetM [17], configuration `3d_fullres` and the default trainer — a ResidualEncoderUNet with six stages and $[32, 64, 128, 256, 320, 320]$ features, sized for a single 16 GB GPU. The fingerprint step resampled volumes to $1 \times 1 \times 1$ mm and set a patch of $112 \times 160 \times 128$ with batch size 2. Training follows the stock schedule (1000 epochs per fold, Dice-plus-cross-entropy with deep supervision, SGD with Nesterov momentum, polynomial decay, standard spatial and intensity augmentation); no custom loss, sampler or augmentation was introduced. All five folds completed on an NVIDIA RTX 4070 Ti SUPER (16 GB). Final predictions come from the five-fold ensemble with mirroring test-time augmentation. Inference is a sliding window whose `step_size` we treat as tunable (Section 3.3). For post-processing, nnU-Net's automatic step selected largest-connected-component (LCC) filtering for labels $\{3, 4, 7\}$ (the two ventricles and left lentiform) and left the rest untouched; we adopt this structure-specific scheme and test it against uniform and milder alternatives. A second recipe merging each left/right pair into a bilateral class (six classes) with mirroring augmentation was only partially trained (one fold) and is reported as future work, not as an evaluated alternative.

3 Results

3.1 Task 1A — Ordinal Artifact Grading

All Task 1A results are computed on leakage-free five-fold OOF predictions over the 532 training planes; no test-set or leaderboard information is used. Table 1 shows the two backbones are individually comparable (≈ 0.827 calibrated) and that ensembling lifts OOF micro-accuracy to 0.839, with anti-overfit calibration adding +0.012 over the naive 0.5 operating point. Adding eight-way TTA on top of the ensemble does not improve the OOF estimate (0.837), so the gain we report comes from the ensemble and the calibration and not from TTA. Per-fold micro-accuracy is 0.821, 0.838, 0.838, 0.843 and 0.859 (mean 0.840, sd 0.013). That spread is the honest context for the rest of this section: differences between configurations of about a point are within the fold-to-fold variation of a 532-image cohort, and we read them as such.

Table 2 breaks the score down by class. Rare classes dominated by true negatives (Banding, Noise, Positioning) achieve the highest three-way accuracy, whereas high-prevalence classes (Motion, Contrast, Distortion) are hardest. The quadratic weighted κ (QWK), which penalises grading errors by ordinal distance, is highest for Noise (0.81) and Motion (0.71) and lowest for Positioning (0.48) and Distortion (0.46) — the artifacts with the least stereotyped spatial signature, and the natural targets for future work.

Detection by acquisition plane. Because the three input channels come from fixed percentile positions, an artifact whose signature is spatially confined may be represented unevenly depending on the orientation in which it was acquired. Table 3 therefore reports positive-case recall — the fraction of truly affected

Table 1. Task 1A configuration ablation on OOF predictions (micro-accuracy over the flattened $\{0, 1, 2\}$ grid). Calibration is the anti-overfit per-class threshold search of Section 2.1. TTA is listed as its own row because it belongs to the submitted predictor but does not improve the out-of-fold estimate. Every value is reproducible from the released OOF predictions and thresholds.

Configuration	@0.5/0.5	Calibrated
EfficientNet-B4 (5 folds)	0.820	0.828
ConvNeXt-Small (5 folds)	0.824	0.826
Ensemble (10 models)	**0.828**	**0.839**
Ensemble (10 models) + TTA×8	0.827	0.837

Table 2. Task 1A per-class OOF performance of the ensemble at calibrated thresholds. n_1, n_2 are mild/severe plane counts; Acc. is three-way accuracy; QWK is quadratic weighted κ over $\{0, 1, 2\}$; Rec.$^+$ is the fraction of truly affected planes (severity ≥ 1) flagged as affected.

Class	n_1	n_2	Acc.	QWK	Rec.$^+$	$(t^{(1)}, t^{(2)})$
Banding	9	12	0.970	0.690	0.476	(0.55, 0.25)
Noise	42	51	0.908	0.813	0.645	(0.65, 0.35)
Positioning	45	25	0.874	0.477	0.357	(0.65, 0.35)
Zipper	142	37	0.812	0.644	0.726	(0.50, 0.50)
Motion	93	76	0.774	0.705	0.527	(0.60, 0.25)
Contrast	166	31	0.771	0.608	0.716	(0.50, 0.40)
Distortion	95	49	0.769	0.459	0.424	(0.65, 0.55)
Mean			**0.839**	**0.628**	**0.591**	—

planes (severity ≥ 1) flagged as affected — per acquisition plane. Detection is markedly orientation-dependent: 0.618 axial, 0.508 coronal and 0.629 sagittal, and of the positives it does detect, the coronal plane assigns the correct severity in 0.362 of cases against 0.565 axial. Per class the pattern follows the geometry of the artifact. Zipper, a discrete band confined to few slices, is recovered in 0.86 of sagittal positives but only 0.45 of coronal ones; banding ranges from 0.17 axial to 0.71 coronal. Positioning and distortion — the classes with the weakest ordinal agreement in Table 2 — stay below 0.52 in every orientation, so their difficulty is representational rather than an orientation effect.

Two caveats keep this from being over-read: prevalence is unbalanced across planes (110 of the 179 zipper positives are sagittal, against 38 axial), so part of each gap reflects how many examples the training folds saw in that orientation; and micro-accuracy runs opposite to recall (0.856 axial, 0.846 coronal, 0.817 sagittal) because true negatives fill most of the grid, which is why recall is the quantity reported here.

What the fixed three-slice input captures. Table 4 isolates the sampling itself. Holding the ten trained models, the fold assignment and the calibrated thresholds fixed, we change only which slices form the three input channels and average predicted probabilities over every triplet a scheme produces; *all slices* slides a

Table 3. Task 1A positive-case recall by acquisition plane (OOF ensemble, calibrated thresholds). n^+ is the number of affected planes of that class in that orientation; Rec.$^+$ is the fraction of them flagged as affected.

	Axial		Coronal		Sagittal	
Class	n^+	Rec.$^+$	n^+	Rec.$^+$	n^+	Rec.$^+$
Noise	32	0.78	24	0.54	37	0.59
Zipper	38	0.55	31	0.45	110	**0.86**
Positioning	16	0.25	34	0.41	20	0.35
Banding	6	0.17	7	0.71	8	0.50
Motion	53	0.40	63	0.68	53	0.47
Contrast	100	0.87	43	0.53	54	0.57
Distortion	40	0.42	44	0.30	60	0.52
All classes	**285**	**0.618**	**246**	**0.508**	**342**	**0.629**

three-slice window across the whole through-plane axis, the densest coverage the trained backbones accept without retraining. The design choice is vindicated in one direction and challenged in another. Collapsing to a replicated central slice costs 0.024 micro-accuracy and drops positive recall from 0.591 to 0.450, so three independent positions earn their place. But exhaustive coverage recovers precisely the artifacts that fixed sampling was suspected of missing: zipper recall rises from 0.726 to 0.799 (axially, 0.55 to 0.74) and positioning from 0.357 to 0.400. The gain is neither free nor general — motion ($0.527 \rightarrow 0.314$) and distortion ($0.424 \rightarrow 0.215$) collapse under dense sampling, consistent with signatures that are inter-slice inconsistencies and are washed out by averaging over strongly overlapping triplets, while specificity falls from 0.947 to 0.940, so net micro-accuracy decreases to 0.826.

The conclusion is deliberately limited: all ten models were *trained* on the 25th/50th/75th-percentile input, so this measures robustness of the learned representation to a change of sampling at inference, not what a densely trained or fully volumetric model would achieve — though the localised-artifact gains are informative precisely because they survive that mismatch. What the experiment rules out is a uniform fix: no single sampling density serves all seven classes, whose evidence lives at different spatial scales.

Table 4. Task 1A slice-sampling ablation, inference only: same ten models, same folds, same thresholds, different input slices. T is the number of slice triplets averaged for a 36-slice stack. The four rightmost columns are positive-case recall for two spatially localised artifacts and two defined by inter-slice inconsistency.

Input slices	T	Micro-acc.	Rec.$^+$	Zip.	Pos.	Mot.	Dist.
Central slice ×3	1	0.816	0.450	0.682	0.314	0.444	0.153
P25/50/75 (adopted)	1	**0.839**	**0.591**	0.726	0.357	**0.527**	**0.424**
5 positions	3	0.834	0.544	0.754	0.300	0.491	0.181
9 positions	7	0.832	0.543	0.704	0.343	0.473	0.208
All slices (sliding)	34	0.826	0.550	**0.799**	**0.400**	0.314	0.215

3.2 Task 1B — Low-Field Enhancement

Table 5 reports the official validation-leaderboard metrics. Among our configurations the native pass-through (C) is best or tied-best on every metric except FID, where it is within one point of the physics denoiser; it uniquely achieves a *negative* BRISQUE change and the lowest FRD. The resampled-target translator (M2) is by far the worst, and the adversarial translator (M3) trades a small CLIP-IQA gain for worse FID and FRD.

Table 5. Task 1B official validation-leaderboard metrics. Arrows indicate the favourable direction; Δ values are changes relative to the input low-field image. Best value *among our submissions* in bold; the top team is shown for context only.

Method	FID↓	LPIPS↓	PSNR↑	ΔBRISQUE↓	ΔCLIP-IQA↑	FRD↓
M1 physics denoiser	130.35	0.467	11.80	+8.64	+0.041	9.35
M2 resampled target	210.46	0.501	11.53	+27.0	+0.084	18.55
M3 unpaired adversarial	136.88	**0.461**	11.93	+7.83	+0.066	9.89
C native pass-through	**131.37**	0.465	**11.95**	−4.76	−0.018	**7.10**
Top team (context)	114.57	0.457	12.07	−0.91	+0.016	10.65

Local proxy fidelity and the Goodhart effect. For methods *not* selected against the local metric, the local-to-server FID ratio is stable (≈ 1.6): the resampled target is correctly ranked far worse than the physics baseline both locally and on the server. The adversarial translator breaks this — selected by local FID, it attains the best local score of all methods (67.3) yet its local-to-server ratio jumps to 2.03 and it is worse than the baseline on the server. This is a textbook Goodhart failure: optimising the proxy detaches it from the target, which is why our M4 selection required held-out FID plus a BRISQUE non-regression guard.

No fixed operator beats the untouched image. On 12 held-out complete subjects the identity (native pass-through) is the FID optimum among fixed operators: unsharp masking, gamma, CLAHE and denoise-then-sharpen all *increase* per-subject FID (from 138.7 up to 151.7) and BRISQUE, so the percentile-normalised native image already sits at a joint optimum.

Registration ceiling and the failure of supervised translation. Table 6 evaluates three ways of bringing CISO onto a low-field grid. Array resizing (the M2 target) yields only 0.36 mean foreground correlation, and affine-header resampling is worse (0.25). Rigid mutual-information registration nearly doubles alignment to 0.63, improving every subject — yet across the full training set the registered correlation averages only 0.45, because cross-field-strength contrast differences and low-field distortion cap rigid registration. Even with correct registration, the M4 perceptual translator fails to beat the control: *every* checkpoint raises held-out FID from 138.7 to 164–173 and ΔBRISQUE to +24 or more, flat across epochs, so at ≈ 0.5 alignment the model synthesises high-field texture at slightly displaced locations, which the reference metrics penalise. Our selection guard correctly rejected all M4 checkpoints.

Table 6. Task 1B low-field-to-CISO alignment (mean foreground correlation). Rigid mutual-information registration nearly doubles alignment over array resizing, but tops out near 0.5–0.6.

Alignment method	Correlation
Array resize (the M2 target)	0.363
Affine-header resample	0.251
Rigid + mutual information (M4)	**0.632**

Preprocessing hygiene. The pass-through control isolates the cost of common preprocessing choices. Our M1 submission applied resize-to-256 and 8-bit quantisation before scoring; replacing this with native resolution and float output moved the server BRISQUE change from +8.6 to −4.8 at essentially unchanged FID. A substantial part of the "enhancement" problem is therefore simply avoiding self-inflicted degradation.

3.3 Task 2 — Subcortical Segmentation

All Task 2 numbers are computed against manual ground truth on the five-fold OOF ensemble over all 79 cases, reported over all eleven annotated labels with the nine-label aggregation of our submission-time reading alongside. Reducing the sliding-window `step_size` from the default 0.5 to 0.3 on a held-out subset improved mean DSC by +0.0042, HD95 by −0.004 mm and ASSD by −0.012 mm, with no case regressing; as the only cost is inference time we adopted 0.3.

A six-way post-processing ablation on the same ensemble makes the inference-tuning point sharply, and Table 7 shows it makes the same point under either label set: the adopted structure-specific scheme (nnU-Net's automatic LCC on labels $\{3, 4, 7\}$) gives the best DSC, HD95 and ASSD under both. Its story is clearest against a uniform policy: extending LCC to every label barely moves DSC ($0.7842 \rightarrow 0.7806$) but inflates ASSD from 0.920 to 1.125 mm — the largest single movement, in the wrong direction. At 0.064 T the thin tails of hippocampus and caudate are often rendered as a large body plus small, barely-attached fragments indistinguishable from noise floaters; uniform LCC deletes the real tail with the noise, and the surface metrics register the amputation immediately. Three milder floater-removal variants ($<$ 30% of largest, $<$ 50 voxels, and a combined rule) confirm that even gentle uniform cleanup does not pay: none beats the adopted scheme on ASSD. Scoring the ventricles makes this check stricter rather than looser, since they are two of the three labels the adopted scheme actually filters; the conclusion survives it unchanged.

Table 8 gives the per-structure breakdown, over all eleven labels (mean Dice 0.7842) and over the nine of our submission-time reading (0.7849); the two differ by less than a thousandth, because the ventricles sit close to the cohort mean. The hippocampus is the bottleneck by a wide margin (both sides near Dice 0.64, worst surface distances and volume errors), while the thalamus is the ceiling (Dice 0.876/0.875). Caudate and lentiform form a consistent middle band around

Table 7. Task 2 post-processing ablation on the five-fold OOF ensemble ($n = 79$), under both label sets. LCC is largest-connected-component filtering. The adopted scheme wins on DSC, HD95 and ASSD under either aggregation; uniform LCC is the clear loser on ASSD under both.

	All 11 labels			9 scored (as submitted)		
Post-processing	DSC↑	HD95↓	ASSD↓	DSC↑	HD95↓	ASSD↓
None (raw ensemble)	0.7836	2.478	0.938	0.7843	2.564	0.999
Per-label (adopted)	**0.7842**	**2.401**	**0.920**	**0.7849**	**2.483**	**0.978**
LCC on all labels	0.7806	2.464	1.125	0.7805	2.560	1.228
Fragments < 30% of largest	0.7835	2.490	1.001	0.7842	2.583	1.074
Fragments < 50 voxels	0.7835	2.491	0.997	0.7842	2.582	1.069
Adopted + < 50 voxels	0.7841	2.416	0.977	0.7848	2.501	1.047

0.80–0.84. The corpus callosum is an instructive outlier: a modest Dice (0.749, as a thin midline sheet) but the best ASSD (0.459 mm) and RVE (7.74 %), since its compact surface is localised precisely even when overlap looks unremarkable. Fig. 1 shows the hippocampal failure mode directly.

Table 8. Task 2 per-structure performance of the five-fold ensemble with the adopted post-processing ($n = 79$). All eleven annotated labels enter the primary mean; the mean over the nine labels of our submission-time reading is given below it for continuity. Ranking under either set puts the same structures at the extremes.

Structure	DSC↑	HD↓	HD95↓	ASSD↓	RVE%↓
Hippocampus L	0.627	5.295	3.653	1.709	22.95
Hippocampus R	0.654	4.974	3.515	1.397	24.03
Ventricle L	0.790	4.786	1.913	0.643	13.55
Ventricle R	0.772	5.155	2.153	0.678	12.70
Caudate L	0.812	4.286	2.319	0.861	14.27
Caudate R	0.802	3.766	2.129	0.818	11.53
Lentiform L	0.837	5.181	2.240	1.024	13.57
Lentiform R	0.833	4.174	2.695	1.101	16.80
Thalamus L	0.876	3.813	2.191	0.763	11.14
Thalamus R	0.875	4.072	2.080	0.666	8.79
Corpus callosum	0.749	4.989	1.520	0.459	7.74
Mean (11 labels)	**0.7842**	**4.590**	**2.401**	**0.920**	**14.28**
Mean (9, as submitted)	*0.7849*	*4.505*	*2.483*	*0.978*	*14.53*

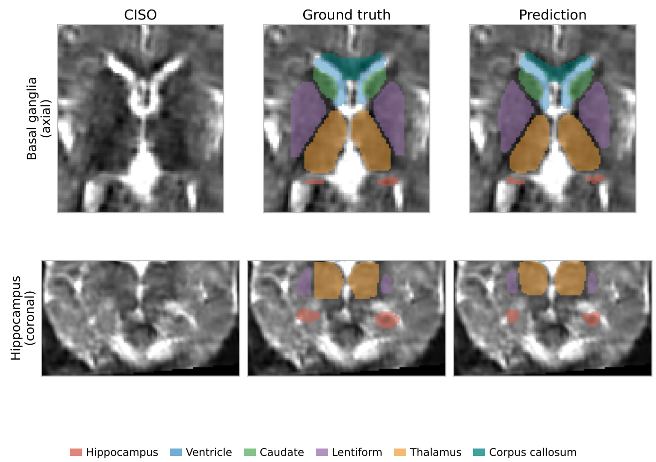

Fig. 1. Task 2 representative segmentation (case at the cohort-median scored Dice, $\approx$ 0.82). Rows: axial through the basal ganglia, coronal through the hippocampus. Columns: CISO volume, ground truth, prediction (ventricles drawn, and scored under the 2026 criteria). Thalamus and lentiform are recovered closely; the hippocampus (red) is visibly under-segmented — the thin-tail, low-contrast failure mode behind its low Dice.

4 Discussion

Across three tasks the same lesson recurs: at 0.064 T a well-configured, honestly validated pipeline is competitive.

Honest evaluation is a method, not an afterthought. In Task 1A, in-fold threshold tuning had earlier produced optimistic estimates that did not transfer; the leakage-free OOF protocol, with a four-of-five-folds acceptance rule for any calibrated threshold, gives an honest 0.839. In Task 2, out-of-fold scoring over the full cohort and a zero-exclusion curation policy keep the numbers faithful to unseen-data behaviour. Task 1B is the same discipline in another guise: a local proxy is trustworthy for *ordering* methods it did not tune, but the adversarial translator selected against that proxy transferred poorly, so we select on held-out data and gate candidates on a metric they were not trained to optimise.

Strong baselines must be beaten, not assumed away. Task 1B's central finding is that a percentile-normalised native pass-through is a joint FID/BRISQUE optimum among simple operations and beats all four learned enhancers on the leaderboard. This is not an argument against learned enhancement but a diagnosis of when it fails here: the input is already near-optimal for the metrics, supervision is capped by a registration ceiling near 0.5–0.6 that makes pixel and feature losses reward texture which is plausible but spatially misplaced, and proxy-driven checkpoint selection invites Goodhart effects. Having no learned parameters, the pass-through also cannot overfit the validation cohort — robustness our learned models lacked. Task 2 carries the mirror-image lesson: connected-component filtering helps only where a dominant component genuinely is the whole structure, and generalising it to small filamentous gray matter deletes real anatomy (ASSD $0.920 \rightarrow 1.125$ mm over all eleven labels), so the right move was to trust nnU-Net's per-label selection. The residual errors are physically interpretable throughout: the diffuse Task 1A Positioning and Distortion artifacts dominate its ordinal error, and the Task 2 hippocampus — small, thin, curved and weakly contrasted at 0.064 T — never closed its early gap, at Dice levels the physics predicts, indicating signal-limited rather than method-limited pipelines.

Where our own design choice does not hold up. The slice-sampling ablation of Table 4 is a negative result about this paper rather than about the field. Three fixed percentile slices clearly beat a single central slice, so the input is not decorative; but they under-detect precisely the artifacts whose signature is spatially confined, and exhaustive coverage recovers much of that loss for zipper and positioning while destroying it for motion and distortion. The remaining headroom in Task 1A therefore sits in the input representation rather than in the architecture — consistent with this paper's thesis, uncomfortable for its own Task 1A design — and no uniform sampling rule can serve seven classes whose evidence lives at different spatial scales. We report it because a baseline paper earns its place only if it also marks where its own baseline is weak.

Limitations and future work. Both cohorts are small and single-source, so our metrics are cross-validation estimates on one release group, not population performance. The most promising next steps are task-specific: better alignment for Task 1B (deformable or contrast-invariant registration aiming for > 0.8 correlation, to test whether supervised enhancement can finally beat the pass-through); a boundary- or topology-aware objective [18] and the full five-fold symmetric-label recipe for the thin-tail hippocampal failures in Task 2; and, for Task 1A, a slice-selection rule conditioned on the artifact class — or a densely sampled, end-to-end volumetric encoder, which our inference-only ablation cannot stand in for — together with higher-resolution features and validation of the label-free threshold transport. All stay within the paper's premise: at 0.064 T the gains come from data, evaluation and inference, not from architecture.

Acknowledgments. Acknowledgments omitted for anonymous review.

Disclosure of Interests. The authors have no competing interests to declare.

References

1. Marques, J.P., Simonis, F.F.J., Webb, A.G.: Low-field MRI: An MR physics perspective. J. Magn. Reson. Imaging **49**(6), 1528–1542 (2019). https://doi.org/10.1002/jmri.26637
2. Sheth, K.N., Mazurek, M.H., Yuen, M.M., et al.: Assessment of brain injury using portable, low-field magnetic resonance imaging at the bedside of critically ill patients. JAMA Neurol. **78**(1), 41–47 (2021). https://doi.org/10.1001/jamaneurol.2020.3263

3. Arnold, T.C., Freeman, C.W., Litt, B., Stein, J.M.: Low-field MRI: clinical promise and challenges. J. Magn. Reson. Imaging **57**(1), 25–44 (2023). https://doi.org/10.1002/jmri.28408
4. Sarracanie, M., Salameh, N.: Low-field MRI: how low can we go? A fresh view on an old debate. Front. Phys. **8**, 172 (2020). https://doi.org/10.3389/fphy.2020.00172
5. Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: Proc. IEEE ICCV, pp. 2980–2988 (2017)
6. Frank, E., Hall, M.: A simple approach to ordinal classification. In: Machine Learning: ECML 2001. LNCS, vol. 2167, pp. 145–156. Springer (2001)
7. Lakshminarayanan, B., Pritzel, A., Blundell, C.: Simple and scalable predictive uncertainty estimation using deep ensembles. In: Advances in Neural Information Processing Systems, vol. 30 (2017)
8. Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., Hochreiter, S.: GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In: Advances in Neural Information Processing Systems, vol. 30 (2017)
9. Zhang, R., Isola, P., Efros, A.A., Shechtman, E., Wang, O.: The unreasonable effectiveness of deep features as a perceptual metric. In: Proc. CVPR, pp. 586–595 (2018). https://doi.org/10.1109/CVPR.2018.00068
10. Mittal, A., Moorthy, A.K., Bovik, A.C.: No-reference image quality assessment in the spatial domain. IEEE Trans. Image Process. **21**(12), 4695–4708 (2012). https://doi.org/10.1109/TIP.2012.2214050
11. Wang, J., Chan, K.C.K., Loy, C.C.: Exploring CLIP for assessing the look and feel of images. In: Proc. AAAI, pp. 2555–2563 (2023). https://doi.org/10.1609/aaai.v37i2.25353
12. Ronneberger, O., Fischer, P., Brox, T.: U-Net: convolutional networks for biomedical image segmentation. In: MICCAI 2015. LNCS, vol. 9351, pp. 234–241. Springer, Cham (2015). https://doi.org/10.1007/978-3-319-24574-4_28
13. Mao, X., Li, Q., Xie, H., Lau, R.Y.K., Wang, Z., Paul Smolley, S.: Least squares generative adversarial networks. In: Proc. ICCV, pp. 2794–2802 (2017). https://doi.org/10.1109/ICCV.2017.304
14. Mattes, D., Haynor, D.R., Vesselle, H., Lewellen, T.K., Eubank, W.: PET-CT image registration in the chest using free-form deformations. IEEE Trans. Med. Imaging **22**(1), 120–128 (2003). https://doi.org/10.1109/TMI.2003.809072
15. Johnson, J., Alahi, A., Fei-Fei, L.: Perceptual losses for real-time style transfer and super-resolution. In: ECCV 2016. LNCS, vol. 9906, pp. 694–711. Springer, Cham (2016). https://doi.org/10.1007/978-3-319-46475-6_43
16. Isensee, F., Jaeger, P.F., Kohl, S.A.A., Petersen, J., Maier-Hein, K.H.: nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. Nat. Methods **18**(2), 203–211 (2021). https://doi.org/10.1038/s41592-020-01008-z
17. Isensee, F., Wald, T., Ulrich, C., Baumgartner, M., Roy, S., Maier-Hein, K.H., Jaeger, P.F.: nnU-Net revisited: a call for rigorous validation in 3D medical image segmentation. In: MICCAI 2024. LNCS, vol. 15009, pp. 488–498. Springer, Cham (2024). https://doi.org/10.1007/978-3-031-72114-4_47
18. Kervadec, H., Bouchtiba, J., Desrosiers, C., Granger, É., Dolz, J., Ben Ayed, I.: Boundary loss for highly unbalanced segmentation. Med. Image Anal. **67**, 101851 (2021). https://doi.org/10.1016/j.media.2020.101851
19. Maier-Hein, L., Reinke, A., Godau, P., et al.: Metrics reloaded: recommendations for image analysis validation. Nat. Methods **21**, 195–212 (2024). https://doi.org/10.1038/s41592-023-02151-z